

Machine Learning Homework 3

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Overview

This report documents solutions and results for three problems. All required figures are embedded directly from the generated PDF outputs in the `code release` directory. Final numerical results are summarized from the printed outputs recorded at the end of each Python file.

1 Problem 1: Polynomial Regression and Ridge Regularization

Setup

We fit a polynomial model of degree 8 to the provided training data, and evaluate on the test set. Let $X \in \mathbb{R}^{n \times 9}$ be the Vandermonde matrix up to degree 8, and $y \in \mathbb{R}^n$ be the targets. The least squares estimator is $\hat{\theta} = (X^\top X)^{-1} X^\top y$. For ridge with regularization parameter $\lambda > 0$, we use $\hat{\theta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$.

Fitted Curve (Unregularized)

Cross-Validation for λ

Effect of λ on Fitted Curve

Key Numerical Results

From the program output:

- Test error (unregularized, part a3): **4.5607**
- Ridge test error: $\lambda = 0.01$: **0.6634** $\lambda = 0.1$: **0.6341** $\lambda = 0.8$: **0.6493** $\lambda = 5$: **0.8207**

2 Problem 2: PCA on ORL Faces

Top Eigenfaces

Reconstruction with Varying k

SNR vs Number of Components

Key Numerical Results

From the program output:

- SNR samples: $k = 20$: **14.53 dB**, $k = 40$: **16.08 dB**, $k = 100$: **18.93 dB**, $k = 200$: **22.69 dB**,
 $k = 300$: **27.40 dB**

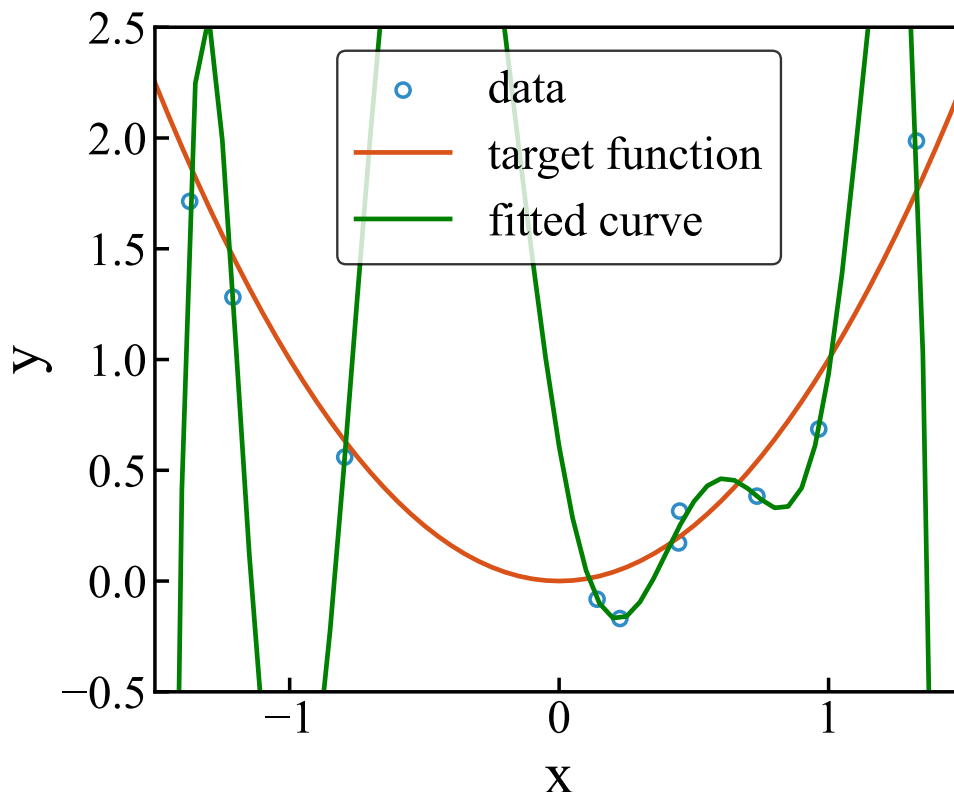


Figure 1: Unregularized degree-8 fit with training data and target function.

- Variance explained: Top 20: **70.60%**, Top 40: **79.23%**, Top 100: **89.10%**, Top 200: **95.38%**, Top 300: **98.43%**

Intuitively, larger k retains more variance, improving reconstruction fidelity and SNR.

3 Problem 3: SVM on MNIST

Part (a): Polynomial Kernels

We train SVMs with polynomial kernels of degrees 1 and 2. The validation error as a function of C is shown below.

Key results from the output:

- Degree 1: Best $C = 46.4159$, Best Val Error: **0.0790**, Test Error: **0.1097**, Test Acc: **0.8903**
- Degree 2: Best $C = 1000.0000$, Best Val Error: **0.0500**, Test Error: **0.0700**, Test Acc: **0.9300**

Part (b): RBF Kernel

We grid-search C and γ ; the validation curves and heatmap are shown below.

Key results from the output:

- Best: $C = 10.0000$, $\gamma = 0.010000$, Best Val Error: **0.0480**
- Test Error: **0.0677**, Test Accuracy: **0.9323**

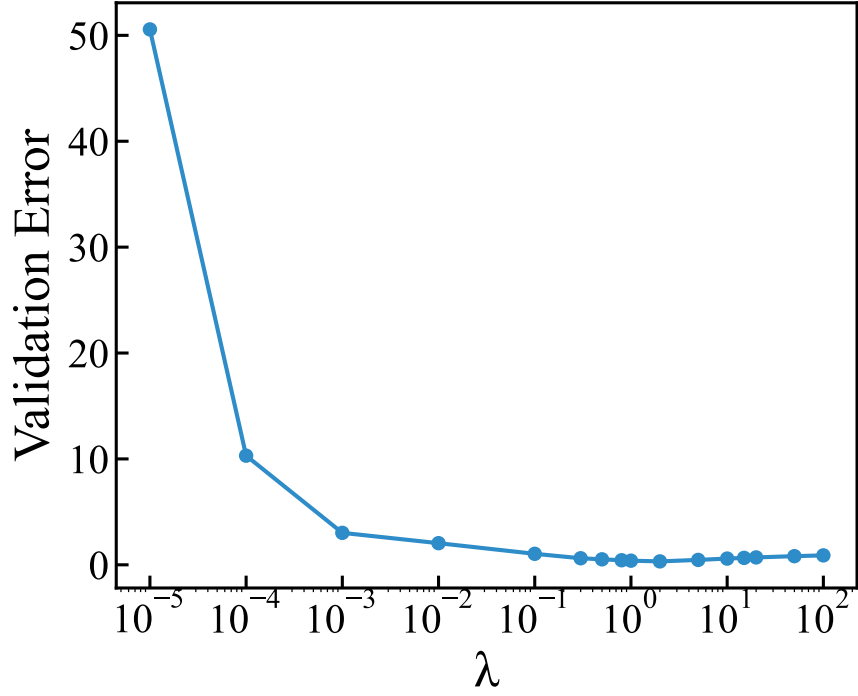


Figure 2: Validation error vs. λ (log-scale).

Conclusion

Across tasks: (1) Ridge regularization substantially reduces overfitting relative to unregularized high-degree polynomial regression; (2) PCA eigenfaces and SNR curves show improved reconstruction with increasing components and quantify the variance retained; (3) On MNIST, polynomial degree 2 and RBF kernels achieve strong validation and test performance, with RBF slightly better on this split.

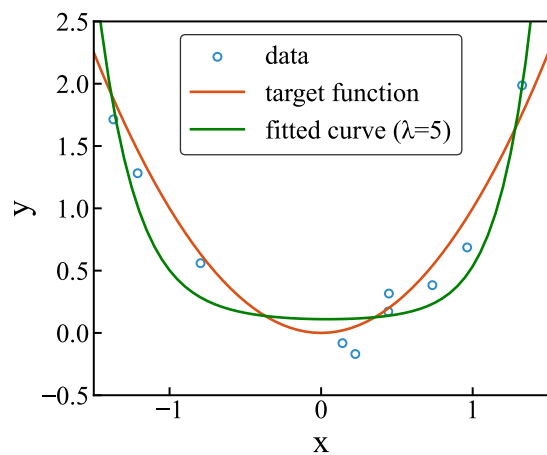
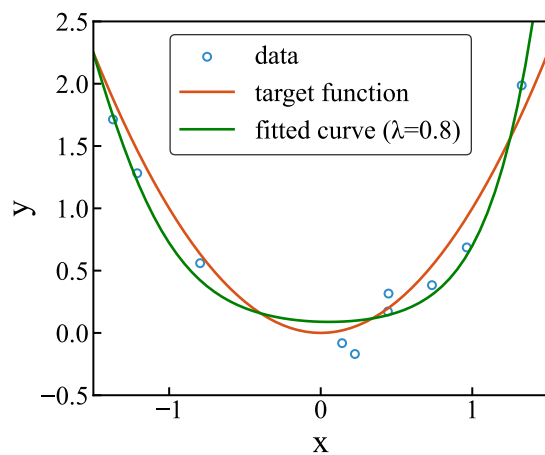
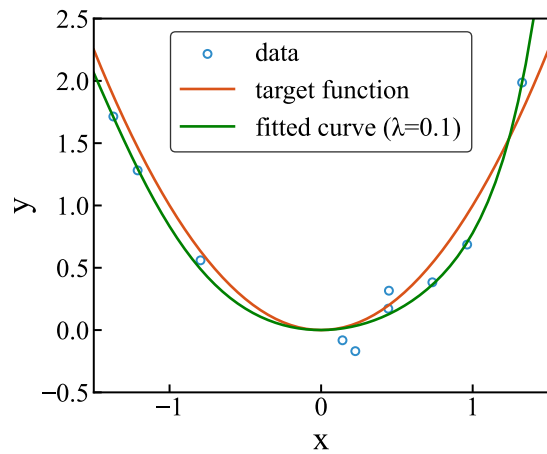
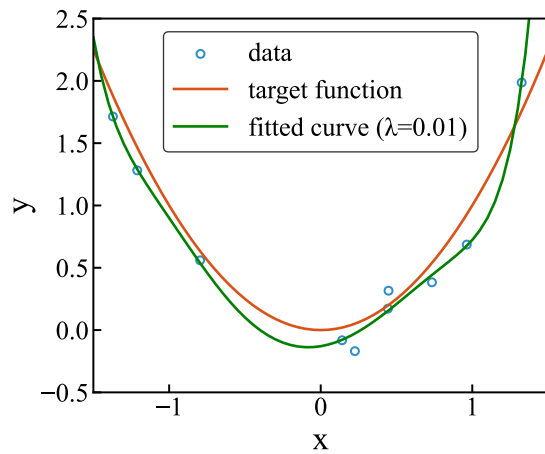


Figure 3: Fitted curves under different λ values.

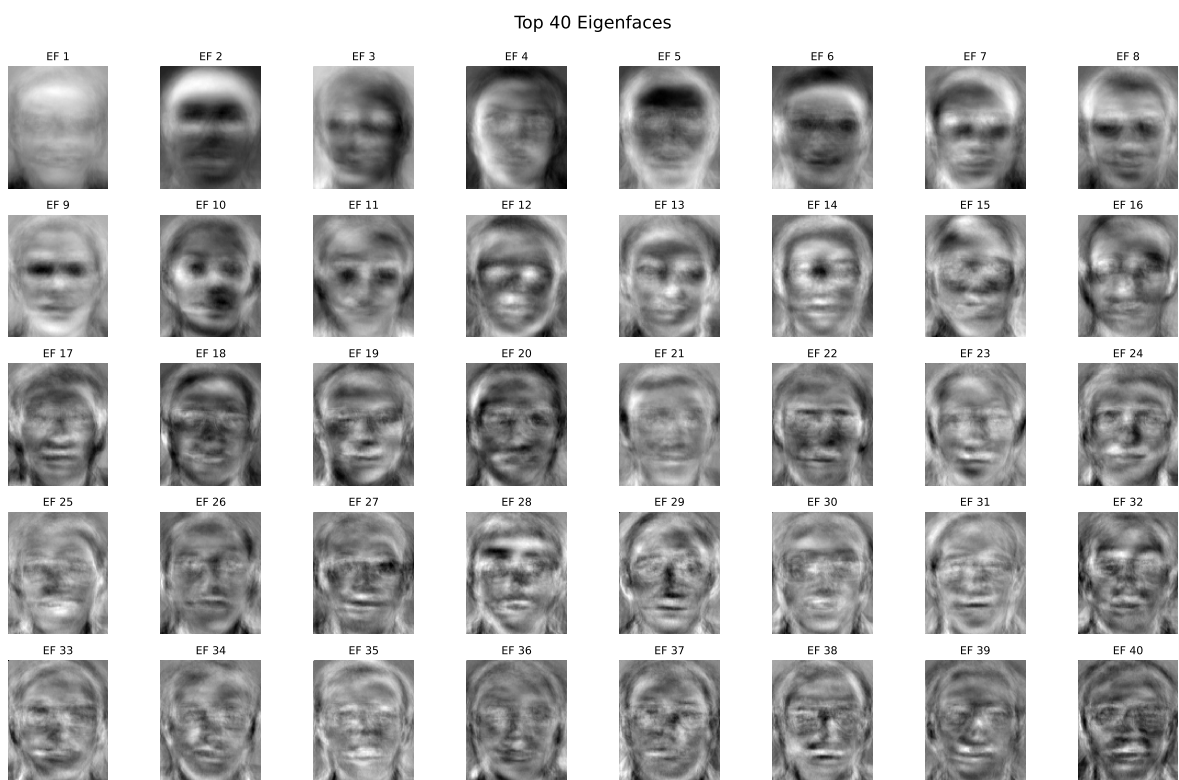


Figure 4: Top 40 eigenfaces learned by PCA.



Figure 5: Reconstruction of selected images with $k \in \{20, 40, 100, 200, 300\}$.

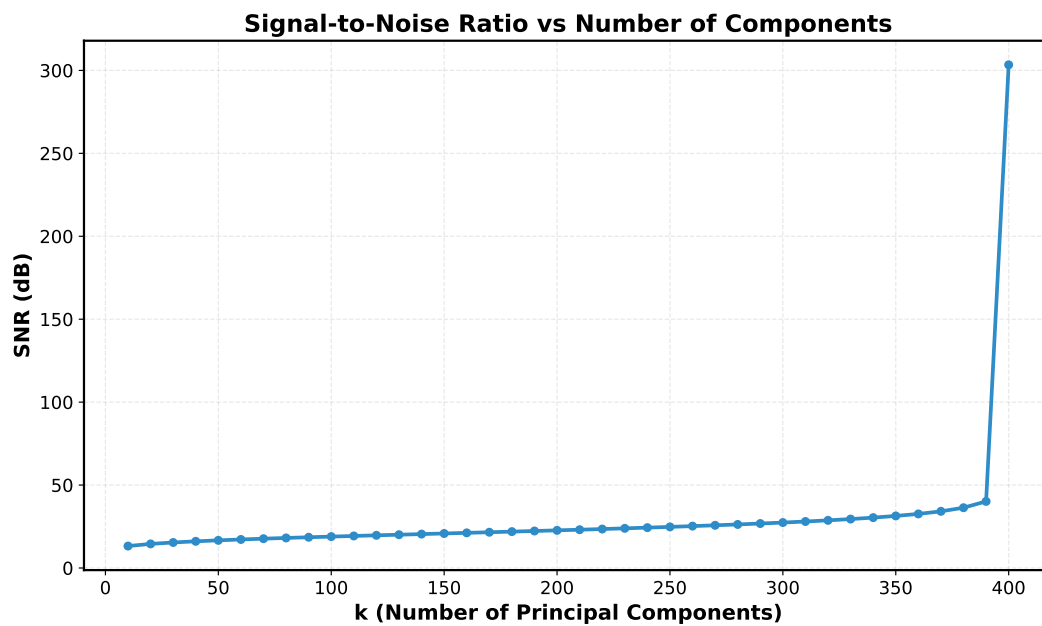


Figure 6: SNR as a function of the number of principal components.

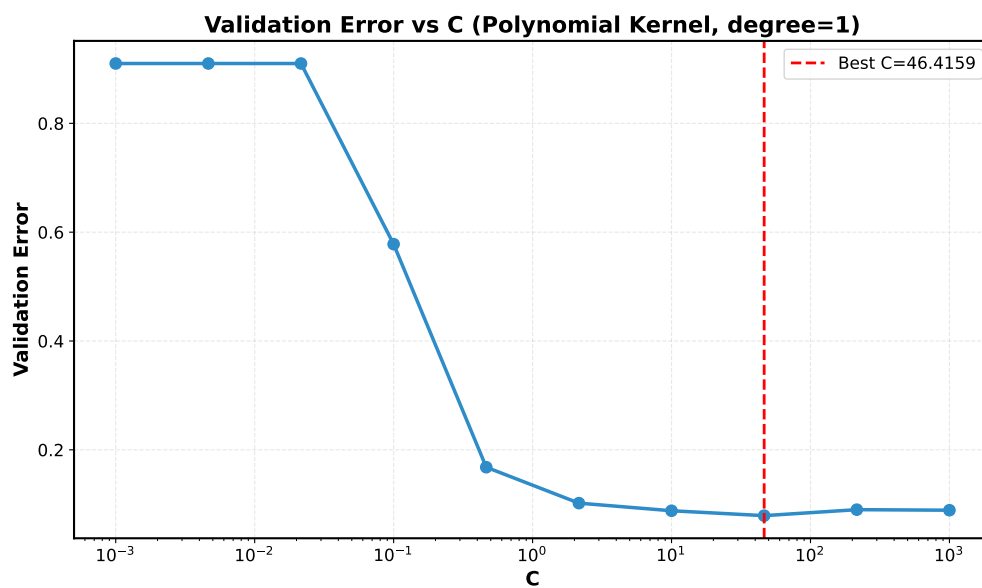


Figure 7: Validation error vs. C for polynomial kernel (degree 1).

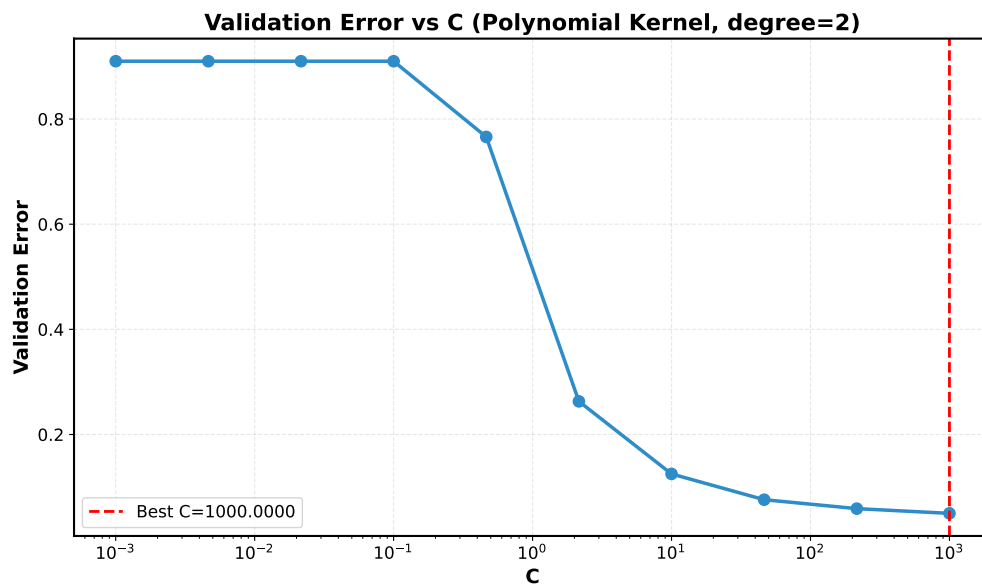


Figure 8: Validation error vs. C for polynomial kernel (degree 2).

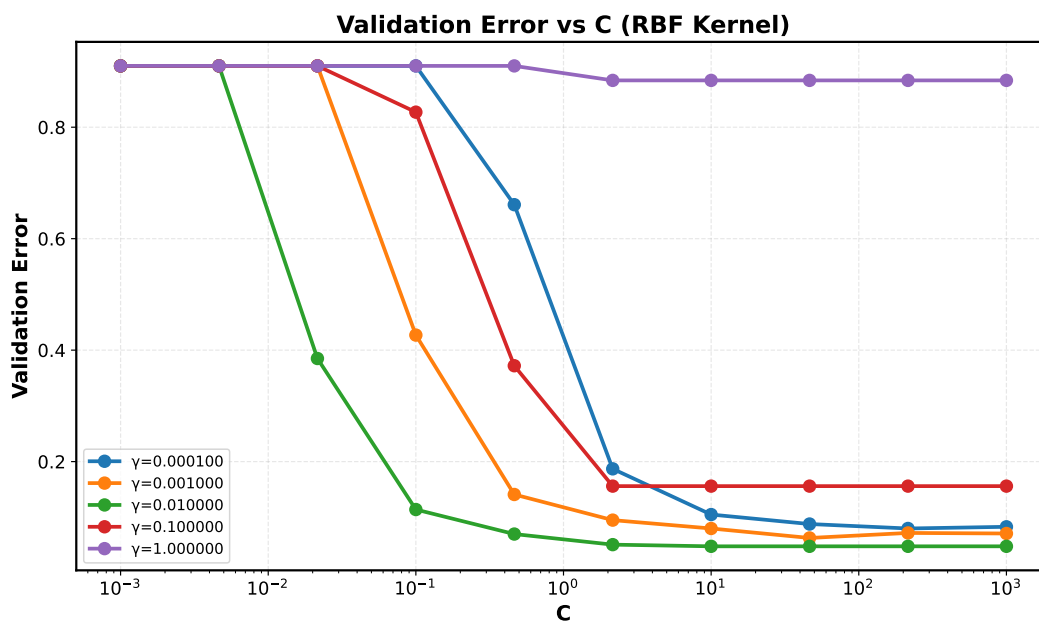


Figure 9: Validation error vs. C across γ values (RBF).

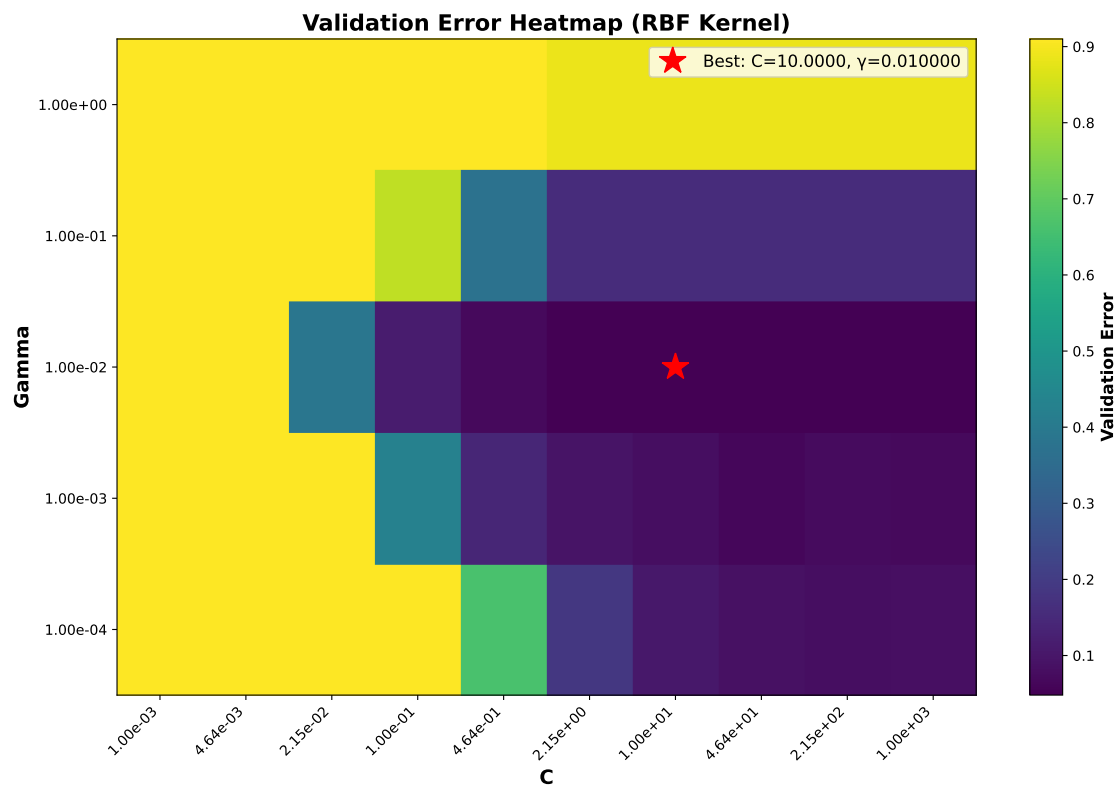


Figure 10: Validation error heatmap over C and γ ; star marks the best.